

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	XXXXXX
Project Name	OptiCrop
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Login System	Secure login and registration functionality	Authentication Module	Done	High
2	Crop Recommendation	Predicts suitable crops using ML algorithms	ML Prediction Module	Done	High
3	Soil Analysis	Analyzes soil parameters like NPK and pH	Analysis Module	Done	Medium
4	Weather Parameter Input	Accepts temperature and humidity inputs	Input Module	Done	Medium
5	Database Management	Stores user and prediction details	Database Module	Done	Medium
6	Input Validation	Validates user input fields	Validation Module	Done	Medium
7	Error Handling	Handles invalid inputs and system errors	Backend Module	Done	Medium
8	Responsive UI	User-friendly responsive interface	Frontend Module	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Responsive pages, Forms
2	Backend / Logic	3	Crop prediction, Validation
3	Database / Storage	1	User data management
4	API / Integration	1	ML model integration
5	Security / Authentication	1	Login authentication

Additional Notes / Comments

The OptiCrop project successfully implements all planned core functionalities with proper modular design and reusable components. The system provides accurate crop recommendations and maintains good usability, validation, and maintainability standards.